

FOOD GROUPS

OBJECTIVES

- 1-Classify food items into different food groups;
- 2-State the major nutrients and group the food items with similar nutrients for food exchange.

Food groups

Definition

A food group consists of a number of food items sharing some common characteristics.

Classification

The two ways of classifying food into groups

1-physiological, on the basis of function

2-on the basis of nutrients

1-Classification Based on Physiological Functions

Food has three basic physiological functions

A-Energy giving

B-Repair and growth

C-Protection and regulation.

Based on the functions, they can be divided into three groups:

Function	Nutrient	Food
Energy giving food	carbohydrates and fats	cereals, sugar , oil
Body building food	proteins	pulses, milk, meat, chicken
Regulatory and protective foods	vitamins and minerals	fruits and vegetables

2-Classification Based on Nutrient

The classification based on the nutrients which they supply.

Five Food Group System

Food group	Foods	Major nutrients present
1- Cereals, Grains and their Products	Rice, wheat, maize, barley, rice flakes, wheat flour etc.	carbohydrates, vitamin B1, vitamin B2, proteins, iron, fiber and folic acid

FOOD GROUPS	FOODS	Major nutrients present
pulses and legumes	Bengal gram, black gram, green gram, red gram lentil ,peas, soya bean	carbohydrates, protein, vitamin B, iron, fiber
Milk	Milk, curd, paneer, cheese, buttermilk, skimmed milk.	proteins, fat, vitamin B2, calcium, vitamin A
Meat and meat products	Meat – chicken, liver, fish, egg	protein, fat, vitamin A, vitamin B2 and vitamin B12

FOOD GROUPS	FOODS	Major nutrients present
Fruits and vegetables	-Mango, guava, orange, papaya, banana, sweet lime, apple .	Vitamin A , vitamin C ,carbohydrates and fiber
	-Vegetables ,pumpkin, Tomatoes and green leafy vegetables like spinach, coriander leaves.	-Carotenoids, vitamin B, folic acid, iron and fiber
Fats and sugar	-Fats: butter, ghee hydrogenated oil, ground nut, mustard, coconut oil.	-Fats
	-Sugars sugar, jaggery, honey	Carbohydrates

CHARACTERISTICS OF FOOD GROUPS

1 -Cereals

1-Whole cereals and grains contain considerable amount of fibre that has a great importance in maintaining a healthy digestive system.

2-Cereals are also rich in vitamin B complex. This gets lost when cereals are refined, So our diet should have whole grains.

3-Eat at least 3 ounces of whole grain breads, cereal, crackers, rice, or pasta each day.

3- Milk

1- it is a rich source of protein, fat, vitamin A and calcium.

2-Get low-fat or fat-free dairy products .

3-People who cannot consume dairy can choose lower lactose or lactose-free dairy products or other calcium sources like (Calcium-fortified juices and soy and rice beverages

3- Meat and meat products

A-(Include meat, poultry, fish)

- 1-Encourage low-fat or lean meats and poultry
- 2- Cooking methods: baking, broiling, grilling
- 3- A rich source of high quality protein, vitamin A and vitamin B.

B – Eggs

- 1- Eggs are a rich source of almost all nutrients except vitamin C.
- 2- Eggs contain proteins of excellent quality therefore they are specifically recommended for growing children , pregnant women and lactating mothers.

C- Pulses and legumes

- 1- major source of protein especially for vegetarians.
- 2- They have a appreciable amount of
vitamin B, calcium and iron.
- 3- protein quality of pulses can be improved by
combining them with cereals.

4-Fruits and Vegetables

Fruits:

- 1-Citrus fruits such as oranges and lemons are rich in vitamin C.
- 2- Yellow fruits like mango, papaya contain carotene which is converted into vitamin A in the body.
- 3-Banana is a good source of iron, calcium and carbohydrates.
- 4-Pomegranate is rich in iron.
- 5- All fruits contain plenty of fiber, which helps in bowel movement.
- 6- Dried fruits like raisins, figs, walnuts and almonds are rich in fats, iron, calcium and fiber.

4-Fruits and Vegetables

Vegetables :

Green leafy vegetables: Green leafy vegetables such as spinach, red and green leaf and romaine and mint are rich in (carotene, calcium, iron and vitamin C).

5 -Fats and sugar

(Butter, ghee, oils like mustard oil, groundnut oil, coconut oil and soya oil).

1- Encourage people to select health-promoting forms of fats:

fat from fish, nuts, and vegetable oils

2-Limit solid fats: butter, stick margarine, shortening, lard,
and visible fat on meat

3-Some amount of fat is needed in daily diet because they supply essential fatty acids. Besides this, some vitamins like

A, D, E and K are fat soluble and important for our body.

FOOD EXCHANGE

The similar food items have been placed together in one food group. Therefore, if we substitute one food for the other in the same group , we will be able to get almost the same nutrients.

Meat group

Protein group



Meat



Egg



Fish

Meats	1 ounce cooked lean beef 1 ounce cooked lean pork or ham
Poultry	1 ounce cooked chicken or turkey, without skin 1 sandwich slice of turkey
Seafood	1 ounce cooked fish or shell fish
Eggs	1 egg

Nuts and seeds	1/2 ounce of nuts (12 almonds, 24 pistachios, 7 walnut halves) 1/2 ounce of seeds (pumpkin, sunflower, or squash seeds, hulled, roasted) 1 Tablespoon of peanut butter or almond butter
Beans and peas	1/4 cup of cooked beans (such as black, kidney, pinto, or white beans) 1/4 cup of cooked peas (such as chickpeas,, lentils, or split peas) 1/4 cup roasted soybeans 1 falafel patty (2 1/4", 4 oz) 2 Tablespoons hummus

Milk group

Milk group



Milk



Cheese



Curd

Milk (choose fat-free or low-fat milk)	1 cup milk
	½ cup evaporated milk
Yogurt (choose fat-free or low-fat yogurt)	1 regular container (8 fluid ounces)
	1 cup yogurt
Cheese (choose reduced-fat or low-fat cheeses)	1 ½ ounces hard cheese (cheddar, mozzarella, Swiss, Parmesan)
	2 ounces processed cheese (American)
	½ cup ricotta cheese
Milk-based desserts (choose fat-free or low-fat types)	1 cup pudding made with milk
	1 cup frozen yogurt
Soy milk (soy beverage)	1 cup calcium-fortified soymilk
	1 half-pint container calcium-fortified soymilk

Vegetables and fruits group

Vegetable and fruit group



Peas



Potatoes



Bananas



Tomatoes

Apple	½ large (3 ¼" diameter) 1 small (2 ¼" diameter) 1 cup, sliced or chopped, raw or cooked
Applesauce	1 cup
Banana	1 cup, sliced 1 large (8" to 9" long)
Cantaloupe	1 cup, diced or melon balls
Grapefruit	1 medium (4" diameter) 1 cup, sections
Mixed fruit (fruit cocktail)	1 cup, diced or sliced, raw or canned
Orange	1 large (3 1/16" diameter)
Peach	1 large (2 ¾" diameter) 2 halves, canned

Pear	1 medium pear (2 ½ per lb) 1 cup, sliced or diced, raw cooked, or canned, drained
Pineapple	1 cup, chunks, sliced or crushed, raw, cooked or canned, drained
Plum	1 cup, sliced raw or cooked 3 medium or 2 large plums
Strawberries	About 8 large berries 1 cup, whole, halved, or sliced, fresh or frozen
Watermelon	1 cup, diced or balls
Dried fruit (raisins, prunes, apricots, etc.)	½ cup dried fruit

Cereals and bread group

Cereal group



Rice



Bread



Chapatis

Breads	1 regular slice 1 small slice, French 4 snack-size slices rye bread
Cornbread	1 small piece (2 ½" x 1 ¼" x 1 ¼")
Crackers	5 whole wheat crackers
English muffins	½ muffin
Muffins	1 small (2 ½" diameter)
Oatmeal	½ cup, cooked

Pancakes	1 pancake (4 ½" diameter) 2 small pancakes (3" diameter)
Popcorn	3 cups, popped
Ready-to eat cereal	1 cup, flakes or rounds 1 ¼ cup, puffed
Rice	½ cup cooked 1 ounce, dry
Pasta--spaghetti, macaroni,	½ cup, cooked
Tortillas	1 small flour tortilla (6" diameter) 1 corn tortilla (6" diameter)

Fat and oil group



Vegetable oils (such as canola, corn, cottonseed, olive, peanut, safflower, soybean, and sunflower)	1 Tbsp
Margarine, soft (trans fat free)	1 Tbsp.
Mayonnaise	1 Tbsp
Olives*, ripe, canned	4 large
Avocado*	½ med
Peanut butter*	2 T
Peanuts, dry roasted*	1 oz
Mixed nuts, dry roasted*	1 oz
Cashews, dry roasted*	1 oz
Almonds, dry roasted*	1 oz
Sunflower seeds*	1 oz

Guidelines for using basic food group :

- 1- Include at least one or a minimum number of serving from each food group in each meal .
- 2- Make choices with each group as foods within each group are similar but not identical in nutritive value.
- 3- If the meal is vegetarian supplement vegetable proteins with suitable combinations to improve the overall protein quality of the diet. For example (cereal, with small quantities of milk).
- 4- Include uncooked vegetables and fruits in the meals.

5- Include at least one serving of milk to ensure a supply of calcium and other nutrients as milk contains all nutrients except iron, vitamin C, and fiber.

6- Cereals should not supply more than 75% of total calories.